

Lecture 13: Tangent Planes & Linear Approximations

Background & Motivation

In Calc I, the tangent line at a point gives the best linear approximation: $f(x) \approx f(a) + f'(a)(x - a)$. Today we extend this to surfaces: the tangent plane is the best flat approximation to $z = f(x, y)$ near a point. This is the foundation of linearization, differentials, and error estimation in multiple dimensions.

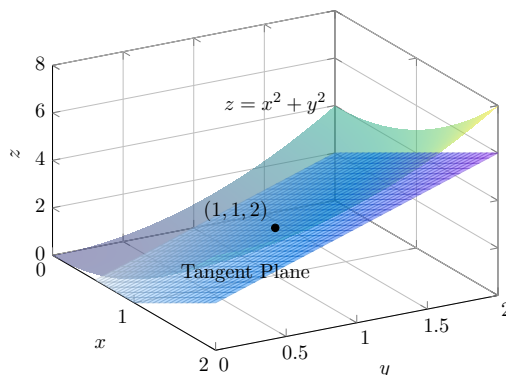
1. From Tangent Lines to Tangent Planes

- In Calc I: tangent line at $x = a$ gives $f(x) \approx f(a) + f'(a)(x - a)$
- In Calc III: the tangent *plane* at (a, b) is the best linear approximation to a surface

Theorem 1: Equation of the Tangent Plane

The tangent plane to $z = f(x, y)$ at the point $(a, b, f(a, b))$:

$$z = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$$



Example 1: Tangent Plane to a Paraboloid

Find the tangent plane to $z = x^2 + y^2$ at $(1, 1, 2)$.

1. **Compute partials:**

$$f_x = 2x, \quad f_y = 2y$$

2. **Evaluate at $(1, 1)$:**

$$f_x(1, 1) = 2, \quad f_y(1, 1) = 2, \quad f(1, 1) = 2$$

3. **Tangent plane:**

$$z = 2 + 2(x - 1) + 2(y - 1) = 2 + 2x - 2 + 2y - 2 = 2x + 2y - 2$$

Example 2: Tangent Plane to $z = y^4 + xy - 2y$ at $(1, 1)$

Find the equation of the tangent plane.

1. **Function value:** $f(1, 1) = 1 + 1 - 2 = 0$

2. **Partials:**

$$f_x = y \implies f_x(1, 1) = 1$$

$$f_y = 4y^3 + x - 2 \implies f_y(1, 1) = 4 + 1 - 2 = 3$$

3. **Result:** $z = 0 + 1(x - 1) + 3(y - 1) = x + 3y - 4$

2. Differentiability

Definition 1: Differentiability in Several Variables

$f(x, y)$ is **differentiable** at (a, b) if:

$$f(a + \Delta x, b + \Delta y) = f(a, b) + f_x \Delta x + f_y \Delta y + E(\Delta x, \Delta y)$$

where the error E satisfies $\frac{E}{\sqrt{(\Delta x)^2 + (\Delta y)^2}} \rightarrow 0$ as $(\Delta x, \Delta y) \rightarrow (0, 0)$.

- If f_x and f_y are **continuous** near (a, b) , then f is differentiable there
- Differentiable $\implies$ continuous (but not the other way around)

3. Linear Approximation

Linear Approximation Formula

$$f(x, y) \approx L(x, y) = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$$

- The tangent plane *is* the linear approximation
- Accuracy improves as (x, y) gets closer to (a, b)

Example 3: Practical Linear Approximation

Use linear approximation to estimate $f(2.1, 0.9)$ where $f(x, y) = x^2y - xy^3$.

1. **Base point:** $(a, b) = (2, 1)$, $f(2, 1) = 4 - 2 = 2$

2. **Partials at base point:**

$$f_x = 2xy - y^3 \implies f_x(2, 1) = 4 - 1 = 3$$

$$f_y = x^2 - 3xy^2 \implies f_y(2, 1) = 4 - 6 = -2$$

3. **Approximate:**

$$L(2.1, 0.9) = 2 + 3(0.1) + (-2)(-0.1) = 2 + 0.3 + 0.2 = 2.5$$

Check: $f(2.1, 0.9) = (4.41)(0.9) - (2.1)(0.729) = 3.969 - 1.531 = 2.438$. Error ≈ 0.06 .

4. Tangent Planes to Implicit Surfaces

– For a surface given implicitly as $F(x, y, z) = k$, the gradient ∇F is normal to the surface

Tangent Plane to $F(x, y, z) = k$ at (x_0, y_0, z_0)

$$F_x(x_0, y_0, z_0)(x - x_0) + F_y(x_0, y_0, z_0)(y - y_0) + F_z(x_0, y_0, z_0)(z - z_0) = 0$$

Equivalently: $\nabla F \cdot \langle x - x_0, y - y_0, z - z_0 \rangle = 0$

Example 4: Tangent Plane to a Sphere

Find the tangent plane to $x^2 + y^2 + z^2 = 14$ at $(1, 2, 3)$.

1. **Set up:** $F(x, y, z) = x^2 + y^2 + z^2$

2. **Gradient:** $\nabla F = \langle 2x, 2y, 2z \rangle$. At $(1, 2, 3)$: $\nabla F = \langle 2, 4, 6 \rangle$

3. **Tangent plane:**

$$2(x - 1) + 4(y - 2) + 6(z - 3) = 0$$

$$2x + 4y + 6z = 28 \implies \boxed{x + 2y + 3z = 14}$$

Practice Problems

Problem 1. Find the tangent plane to $z = xe^y$ at $(2, 0, 2)$.

$$f_x = e^y \implies f_x(2, 0) = 1. \quad f_y = xe^y \implies f_y(2, 0) = 2.$$

$$z = 2 + 1(x - 2) + 2(y - 0) = x + 2y.$$

Answer: $z = x + 2y$

Problem 2. Use linear approximation to estimate $\sqrt{(3.02)^2 + (3.97)^2}$.

Hint: Let $f(x, y) = \sqrt{x^2 + y^2}$ with base point $(3, 4)$.

$$f(3, 4) = 5. \quad f_x = x/\sqrt{x^2 + y^2} = 3/5, \quad f_y = y/\sqrt{x^2 + y^2} = 4/5.$$

$$L(3.02, 3.97) = 5 + \frac{3}{5}(0.02) + \frac{4}{5}(-0.03) = 5 + 0.012 - 0.024 = 4.988.$$

Answer: 4.988

Problem 3. Find the tangent plane to $x^2 + 2y^2 + 3z^2 = 36$ at $(1, 2, 3)$.

$F(x, y, z) = x^2 + 2y^2 + 3z^2$. $\nabla F = \langle 2x, 4y, 6z \rangle$. At $(1, 2, 3)$: $\nabla F = \langle 2, 8, 18 \rangle$.

$$2(x - 1) + 8(y - 2) + 18(z - 3) = 0 \implies x + 4y + 9z = 36.$$

Answer: $x + 4y + 9z = 36$

Problem 4. Find the tangent plane to $z = \sin(xy)$ at $(\pi/2, 1, 1)$ and use it to approximate $f(1.5, 1.1)$.

$$f_x = y \cos(xy) \implies f_x(\pi/2, 1) = \cos(\pi/2) = 0$$

$$f_y = x \cos(xy) \implies f_y(\pi/2, 1) = \frac{\pi}{2} \cos(\pi/2) = 0$$

Tangent plane: $z = 1$. So $f(1.5, 1.1) \approx 1$.

Both partials vanish — the surface is flat at this peak.

Answer: tangent plane $z = 1$, $f(1.5, 1.1) \approx 1$

Problem 5. Two surfaces $z = x^2 + y^2$ and $z = 4 - x^2 - y^2$ intersect along a curve. Find the tangent line to this curve at $(1, 1, 2)$.

Tangent plane to $z_1 = x^2 + y^2$: $z = 2 + 2(x - 1) + 2(y - 1)$, i.e., $2x + 2y - z = 2$.

Tangent plane to $z_2 = 4 - x^2 - y^2$: $z = 2 - 2(x - 1) - 2(y - 1)$, i.e., $2x + 2y + z = 6$.

Tangent line direction: $\langle 2, 2, -1 \rangle \times \langle 2, 2, 1 \rangle = \langle 4, -4, 0 \rangle \propto \langle 1, -1, 0 \rangle$.

Tangent line: $\mathbf{r}(t) = \langle 1 + t, 1 - t, 2 \rangle$.

Answer: $\mathbf{r}(t) = \langle 1 + t, 1 - t, 2 \rangle$